

Research Synthesis & Upgrade Plan

AI-Driven Marp/Markdown Presentation Workflow

Agenda

Today's Topics

1. Design Best Practices

Mayer's principles, typography, layout

2. Color, Accessibility & Data Viz

WCAG, palettes, chart frameworks

3. Narrative & Anti-Patterns

SCQA, slide count, F1-F8

4. AI Tool Landscape

6 tools surveyed, key gap found

5. Open-Source Pipelines

sunbigfly, Akxan, DeepSlide

6. Vision QA & Upgrade Roadmap

VLM contracts, 4-phase plan

Design Best Practices

- **Cognitive Load Theory** – Mayer's 12 principles, validated across 200+ experiments
- **Coherence** – every element must serve the message, nothing extraneous
- **Signaling** – use contrast and bold for key takeaways, never font size alone
- **Segmenting** – max 5 +/- 2 items per slide, progressive disclosure
- **Pre-training** – define key terminology upfront in title and outline



Typography & Visual Hierarchy

- **7-level scale:** 48pt > 40pt > 28pt > 22pt > 18pt > 14pt > 12pt
- **Line height:** body 1.5x, headings 1.1x (unitless CSS)
- **Hard rules:** body $\geq 14\text{px}$ and $\leq 24\text{px}$, max 3 font families
- **7 bento templates:** full-bleed, 2-col, 3-col, 4-card, 6-card, center



Color, Accessibility & Data Viz

4.5:1

WCAG AA body contrast

3:1

Large text minimum

5

Max colors per slide

Principle: Pair icons, patterns, or labels with color. Never rely on color alone to convey information (WCAG 2.1 SC 1.4.1).



Palette

Base > Surface > Primary > Accent > Text



Best Charts

Bar, line, scatter, stacked bar



Avoid

Pie >3 segments, 3D, dense tables

Narrative Structure & Failure Modes

SCQA

Situation > Complication > Question > Answer

1:2

Pacing ratio – 1 slide per 2 minutes, 15 min = 7-8 slides

F2

Overfill

Reduce font or count

F6

Contrast Failure

Adjust colors per WCAG

F8

Narrative Disconnect

Restructure with SCQA

AI Presentation Tool Landscape

Tool	Layout Engine	Quality Mechanism
Gamma.app	Template-constrained	Human review, no visual QA
Beautiful.ai	Constraint satisfaction	Template design rules only
Tome	Narrative-first	No pixel-level QA
Canva Magic	Score heuristics	Layout balance ranking
MS Copilot	Template-constrained	Template guardrails
Pitch	Design system	Human review only

Key gap: No major tool uses vision feedback to refine slide appearance

Open-Source Agentic Pipelines



sunbigfly/ppt-agent-skills

Multi-agent state machine, 4 subagents, JSON contract architecture, Pillow pixel QA, stateless resume



Akxan/ppt-agent-skill

26 production styles, 14 typography rules, 18 chart types (SVG), 8 failure modes with fix protocols



DeepSlide (arXiv)

Time-budgeted planner, Markov style inheritance, dual-scoreboard evaluation (artifact + delivery)

Common Gaps & Best Format

VLM Gap

No open-source tool uses vision-language models for semantic slide QA. Analysis is purely pixel-statistical.

BEST FORMAT

HTML/CSS as primary target, convert to Marp markdown

DEEPSLIDE INNOVATION

Logical-chain planner with time budgets, style inheritance, dual-scoreboard evaluation

Vision QA & Style Contracts

- Stage A:** per-slide visual audit – balance, readability, color
- Stage B:** deck-wide narrative audit – arc, progression, pacing
- Use VLM for **semantic** checks, CSS/DOM for **quantitative**
- theme-contract.json:** full design system encoding per deck



Upgrade Roadmap

Phase 1 – Foundation

Style contract system,
CSS/DOM inspection QA,
density contract validator

WEEK 1

Phase 2 – Pipeline

5-stage pipeline shell, per-
slide JSON contracts,
stateless resume support

WEEK 2

Phase 3 – Quality

VLM QA pass with structured
prompts, feedback loop,
chart MCP tool

WEEK 3

Phase 4 – Polish

Icon search MCP tool,
narrative arc planning,
integration testing

WEEK 4